



Indicaciones: Cada hoja entregada debe indicar nombre, número de C.I. y número de hoja.
Complete el siguiente cuadro y entréguelo junto con el parcial.

Nombre		C.I.	
Número		Carrera	

Preguntas	1	2	3	Total
Número de hojas por preguntas				

***** Cada pregunta se deberá comenzar en una hoja nueva. *****

Factores de conversión e información adicional

$$1 \text{ inch} = 0.0833 \text{ ft} = 2.54 \text{ cm}$$

$$1 \text{ lb}^*/\text{in}^2 = 1 \text{ psia} = 0.06803 \text{ atm}$$

$$1 \text{ atm} = 101325 \text{ Pa}$$

$$1 \text{ kg}/\text{m}^3 = 0.0624 \text{ lb}/\text{ft}^3$$

$$1 \text{ hp} = 550 \text{ lb}^* \text{ ft}/\text{s}$$

$$1 \text{ hp} = 745.7 \text{ W}$$

$$1 \text{ lb} = 0.454 \text{ kg}$$

$$1 \text{ cP} = 10^{-3} \text{ kg}/(\text{m s}) = 6.72 \times 10^{-4} \text{ lb}/(\text{ft s})$$

$$g = 9.8 \text{ m}/\text{s}^2 = 32.2 \text{ ft}/\text{s}^2$$

$$g_c = 32.2 \text{ lb ft}/(\text{lb}^* \text{ s}^2)$$

$$1 \text{ L}/\text{min} = 0.000589 \text{ ft}^3/\text{s}$$

$$1 \text{ ft}^3/\text{s} = 448.8 \text{ gpm}$$

$$1 \text{ m}^3/\text{s} = 15850 \text{ gpm}$$

$$1 \text{ bar} = 1 \times 10^5 \text{ Pa}$$

Tabla de primitivas

$$\int \sqrt{a+bx} \, dx = \frac{2}{3b} (a+bx)^{3/2}$$

$$\int \frac{x^2 \, dx}{\sqrt{a+bx}} = \frac{2(8a^2 - 4abx + 3b^2x^2)}{15b^3} \sqrt{a+bx}$$

$$\int x \sqrt{a+bx} \, dx = \frac{2(3bx - 2a)(a+bx)^{3/2}}{15b^2}$$

$$\int \frac{x \, dx}{\sqrt{a+bx}} = \frac{2(bx - 2a)}{3b^2} \sqrt{a+bx}$$

$$\int x^2 \sqrt{a+bx} \, dx = \frac{2(8a^2 - 12abx + 15b^2x^2)(a+bx)^{3/2}}{105b^3}$$

$$\int \frac{dx}{\sqrt{a+bx}} = \frac{2\sqrt{a+bx}}{b}$$

$$\int \frac{\sqrt{a+bx}}{x} \, dx = 2\sqrt{a+bx} + a \int \frac{dx}{x\sqrt{a+bx}}$$

$$\int \frac{dx}{x^2 \sqrt{a+bx}} = -\frac{\sqrt{a+bx}}{ax} - \frac{b}{2a} \int \frac{dx}{x\sqrt{a+bx}}$$

$$\int \frac{dx}{x\sqrt{a+bx}} = \frac{1}{\sqrt{a}} \log \frac{\sqrt{a+bx} - \sqrt{a}}{\sqrt{a+bx} + \sqrt{a}} \quad (a \text{ pos.})$$

$$\int \frac{dx}{x\sqrt{a+bx}} = \frac{2}{\sqrt{-a}} \tan^{-1} \sqrt{\frac{a+bx}{-a}} \quad (a \text{ neg.})$$

CONSULTA AL TRIBUNAL (Recorte la tirilla por la línea superior y entréguela a un docente)

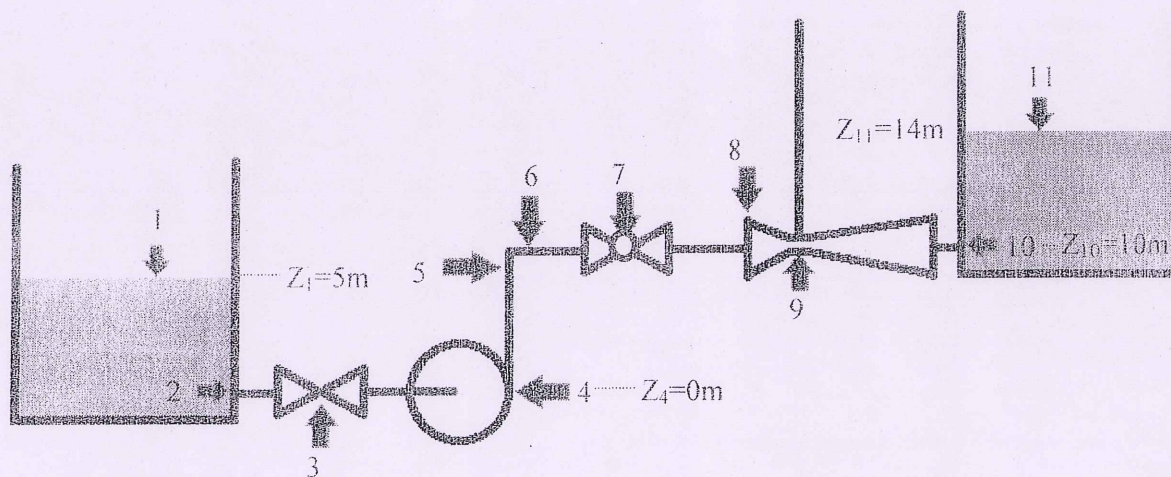
Nombre:

Pregunta:

Nombre:

Pregunta:

- 1) El circuito de la figura se emplea para airear una solución acuosa durante su bombeo. Se utiliza la bomba cuya tabla y ecuación de ajuste se adjuntan. La cañería es de acero comercial Sch 40 de diámetro nominal $1\frac{1}{2}$ ", tiene dos tramos horizontales y un tramo vertical, y totaliza una longitud de 20m. Los accesorios son una válvula esclusa (en succión), una válvula globo (en descarga), un codo 90° estándar, un venturi y conexiones a tanques con resalte al interior. El venturi se utiliza para aspirar aire atmosférico mediante la generación de vacío en su garganta. Este venturi tiene un diámetro mayor idéntico al de la cañería y un diámetro de garganta de $\frac{1}{2}$ ", y conectada a ella se encuentra una tubería vertical de diámetro nominal $\frac{1}{4}$ " (acero comercial Sch 10) y de 6m de largo, abierta a la atmósfera en su extremo superior.
- Determine la pérdida de carga en el venturi, expresada en altura de fluido, cuando su garganta se encuentra a presión atmosférica.
 - ¿Cuál sería la cota mínima de velocidad de giro de la bomba (en rpm), requerida para que exista aspiración de aire en la garganta del venturi?
 - ¿Cuál sería el consumo diario de energía eléctrica en kW.h si se operara en la velocidad correspondiente a la cota mínima?
 - Grafique cualitativamente sobre un único par de ejes, la presión estática en función de la distancia recorrida por el fluido, entre el punto 1 y el punto 11, indicando en el eje de las abscisas los puntos numerados, para las siguientes situaciones:
 - Bomba a velocidad nominal y válvula globo parcialmente cerrada, para obtener el caudal correspondiente a la parte a).
 - Bomba a velocidad ajustada, de modo que se logra el mismo caudal que en la parte anterior pero con la válvula globo completamente abierta.



La longitud de cañería entre la descarga del venturi y el tanque es despreciable.

El tanque contiene la solución a 20°C y está abierto a la atmósfera.

Propiedades de la solución a 20°C : $\mu = 2 \text{ cP}$; $\rho = 1000 \text{ kg/m}^3$; $P_{\text{vap}} = 0.3388 \text{ psi}$



Rugosidad del acero comercial: $4,57 \times 10^{-5}$ m

Propiedades del aire atmosférico:

$P = 1,0$ bar; $T = 15^\circ\text{C}$; $C_p/C_v = 1,4$; $PM = 28,8\text{g/mol}$ $\mu = 1,8 \times 10^{-5}$ Pa.s

Pérdida de carga en el venturi: 15% de la diferencia entre P_8 y P_9 (puede asumirse que dicha pérdida de carga se produce en la zona divergente)

La eficiencia del motor puede asumirse constante e igual a 85% y las pérdidas en el variador de velocidad se pueden despreciar. P_m es la potencia entregada al eje de la bomba.

$Q(\text{m}^3/\text{h})$	2,5	5,0	7,5	10,0	12,5	15,0	17,5	20,0	22,5	25,0
$H(\text{m})$	17,9	17,7	17,2	16,6	15,8	14,7	13,5	12,2	10,6	8,83
$P_m(\text{kW})$	0,28	0,53	0,76	0,96	1,12	1,24	1,32	1,35	1,38	1,4

Curva de ajuste a 2900rpm: $H(\text{m}) = 18 + 8 \times 10^{-3} \cdot Q(\text{m}^3/\text{h}) - 1,5 \times 10^{-2} \cdot (Q(\text{m}^3/\text{h}))^2$

Comience en una hoja nueva el siguiente problema

2) Para el mismo sistema del problema anterior:

- Determine con la mayor exactitud posible cuál debería ser el vacío en el punto 9 para que ingresen al venturi 240L de aire atmosférico por minuto. Justifique el modelo empleado y detalle la ruta de cálculo.
- Estime cuál es la presión máxima en la garganta del venturi que permitiría alcanzar velocidad sónica en el flujo de aire.
- ¿Cuál sería la máxima velocidad del flujo de aire en la condición de la parte b)?



Comience en una hoja nueva el siguiente problema

3) El esquema representa una línea de alimentación de vapor a un intercambiador de calor y parte del circuito de retorno de condensado. El material utilizado en el tendido del sistema es acero al carbono ASTM A 106. A pocos meses de instalado se han producido roturas en el codo (señalado con el círculo punteado) y movimientos en soportes.

a) ¿Puede haber una explicación al rápido deterioro observado?

b) Se plantean como soluciones al problema:

(i) Aumentar el diámetro de la tubería de retorno de condensado a 2,5 in

(ii) Disminuir el diámetro de retorno de condensado a 1 in

(iii) Cambiar el material de la tubería de retorno de condensado por uno menos rígido, polipropileno unido por roscado.

Analice las alternativas planteadas

Datos en la salida del intercambiador:

Presión : 1,5 barg

Diámetro de tubería: 2 in

Flujo de condensado: 10 lb/s

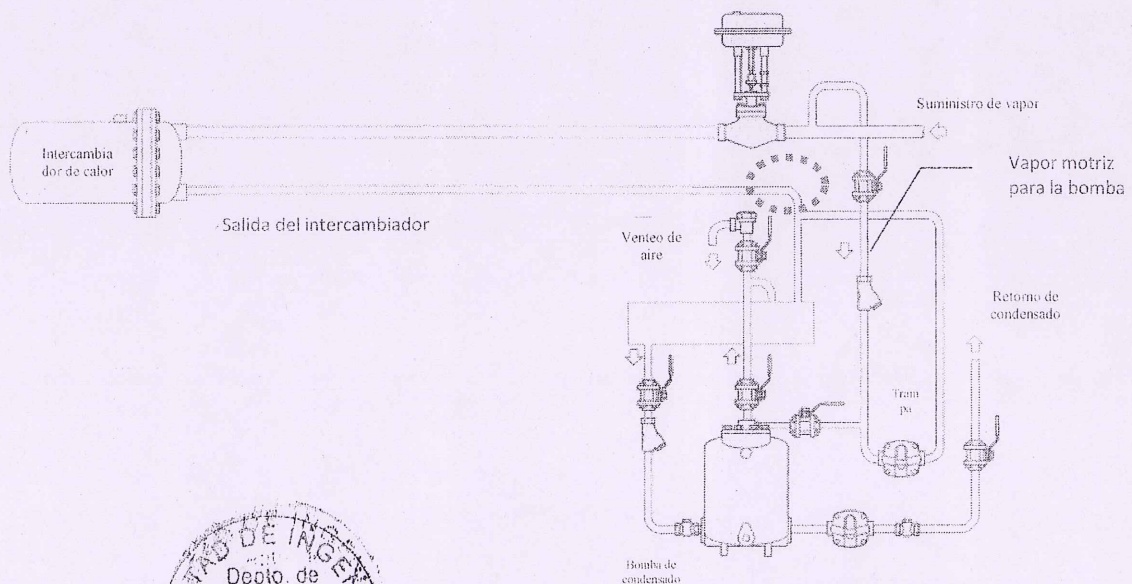
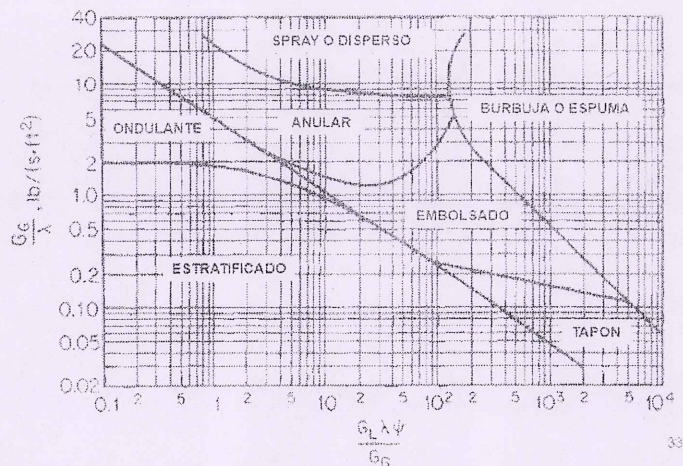
Flujo de vapor: 0,03 lb/s

Diagrama de Baker

$\lambda: 1$ $\psi: 0,6$

Nota: a efectos de simplificar el análisis suponga despreciables los efectos de erosión.

Diagrama de Baker



$$1) \textcircled{a} P_9 = P_{atm} \quad P_{10} = P_{atm} + 4 p_8$$

$$\Delta h_{fv} = 0,15 \left(\frac{P_8 - P_9}{\rho g} \right) \Rightarrow \frac{P_8}{\rho g} = \frac{\Delta h_{fv}}{0,15} + \frac{P_9}{\rho g}$$

$$\text{BEM } 8 \rightarrow 10 \quad \left. \begin{aligned} \frac{\Delta u^2}{2g} + \frac{\Delta z}{g} + \Delta h_{fv} + \frac{\Delta P}{\rho g} &= 0 \\ 0 & \quad 0 \end{aligned} \right\} \Rightarrow \Delta h_{fv} = \frac{\Delta h_{fv} + \frac{P_9}{\rho g} - \frac{P_{10}}{\rho g}}{0,15}$$

$$\Rightarrow \Delta h_{fv} = \frac{P_8 - P_{10}}{\rho g}$$

$$\Rightarrow \Delta h_{fv} = \left(\frac{P_9 - P_{10}}{\rho g} \right) \left(\frac{1}{1 - \frac{1}{0,15}} \right) = 4 \cdot 0,1765 \Rightarrow \boxed{\Delta h_{fv} = 0,71 \text{ m}}$$

6) Condición límite $P_9 = P_{atm} \Rightarrow$ las condiciones son las de la parte a)

$$\text{BEM } 1 \rightarrow 11: \quad \frac{\Delta u^2}{2g} + \frac{\Delta z}{g} + \frac{\Delta P}{\rho g} + \Delta h_f = H$$

$$\Delta h_f = \left(f_3 \frac{L}{D} + \sum K \right) \frac{u^2}{2g} + \underbrace{\Delta h_{fv}}_{0,71 \text{ m}}$$

$$\text{Necesito } u: \text{ BEM } 8 \rightarrow 9 \quad \frac{\Delta u^2}{2g} + \frac{\Delta P}{\rho g} + \frac{\Delta z}{g} + \Delta h_f = 0$$

$$\Rightarrow \frac{u_9^2 - u_8^2}{2g} = \frac{P_8 - P_{atm}}{\rho g} \quad \left. \begin{aligned} 0 & \quad 0 \end{aligned} \right\} \Rightarrow \frac{u_9^2 - u_8^2}{2g} = \frac{\Delta h_{fv} + 4}{4,71 \text{ m}}$$

$$\Delta h_{fv} = \frac{P_8 - P_{10}}{\rho g} \Rightarrow \Delta h_{fv} = \frac{P_8 - P_{atm} - 4 p_8}{\rho g}$$

$$\text{Por continuidad } u_9 A_9 = u_8 A_8 \Rightarrow 1,267 \times 10^{-4} u_9 = 1,31 \times 10^{-3} u_8$$

$$\Rightarrow u_9 = 10,37 u_8 \Rightarrow \frac{(10,37 u_8)^2 - u_8^2}{2g} = 4,71 \Rightarrow u_8 = \sqrt{\frac{2g \cdot 4,71}{106,5}}$$

$$\Rightarrow u_8 = 0,827 \text{ m/s} = u \quad \left(Q = 4,38 \text{ m}^3/\text{s}, \quad \frac{u^2}{2g} = 0,044 \right)$$

$$Re = \frac{\rho u D}{\mu} = 1,8 \times 10^4 \quad \left. \begin{aligned} e/D &= 1,1 \times 10^{-3} \\ L/D &= 489 \end{aligned} \right\} \Rightarrow f_3 \approx 0,029 \Rightarrow f_3 \frac{L}{D} = 14,18$$

$$\sum K = (340 + 30 + 8) f_r + 1,78 = 9,72 \quad \left. \begin{aligned} 0,021 \end{aligned} \right\} \Rightarrow f_3 \frac{L}{D} + \sum K = 23,9$$

$$\Rightarrow \Delta h_f = 1,05 + 0,71 \Rightarrow \Delta h_f = 1,76 \text{ m} \Rightarrow H = 10,76 \text{ m}$$

$$\Rightarrow \text{La curva de la bomba } \approx \text{N}^{\text{min}} \text{ debe pasar por } Q = 4,38 \text{ m}^3/\text{s} \\ H = 10,76 \text{ m}$$



Table 2. Bed parameters

Description	Value
Bed voidage	0.51
D_v , particle diameter by volume	0.04259 m
Solid pellet density	3127 kg/m ³
Tank-to-particle ratio	10.4

The packed bed support plate and the restriction created by the insulation at the exit and entrance of the bed introduce an additional pressure drop. To estimate this value, the pressure drop was measured with an empty bed for different velocities. A third-order polynomial best-fit equation was then used to calculate the additional pressure loss and subtract it from the total pressure drop. In order to test whether there was flow maldistribution due to the inlet geometry, velocity was measured at various points in the cross section of the first five to ten centimeters of the bed with a vane anemometer and found to be acceptable.

3. Results

Tests were conducted for particle Reynolds numbers between $353 \leq Re_p \leq 5206$. The particle Reynolds number was calculated by using the equivalent particle diameter by volume, D_v . The results of the pressure measurements as a function of air mass flux are provided in Fig. 2. An error propagation uncertainty analysis was conducted and the pressure gradient ($\Delta P/L$) uncertainties are included in the figure. The air mass flux (kg/m²-s) uncertainty ranged from 3.1% to 3.8% and was not included in order to maintain clarity within the plot.

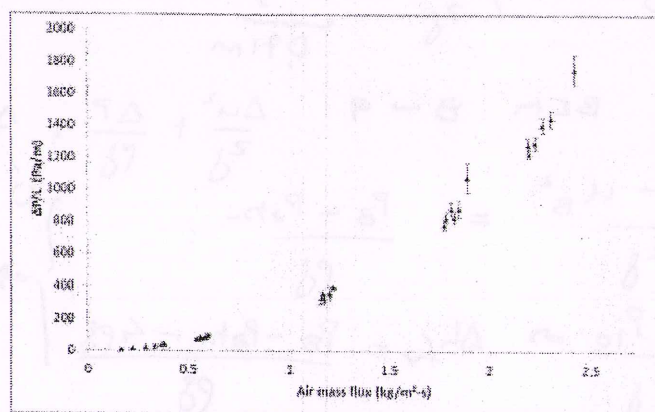


Fig. 2. Measured pressure gradient versus air mass flux.

Since the particle sphericity is difficult to calculate with irregular shaped solids, it was deduced from Ergun's correlation, Eqn. (11), by calculating the root mean square deviation between measured and predicted pressure drop for different sphericity values. This was repeated with Singh et al.'s equation using data that fell in the particle Reynolds number range that was used to develop the correlation, i.e. between $1047 \leq Re_p \leq 2674$. The same sphericity value of 0.495 was obtained using both correlations.

Figure 3 demonstrates the measured pressure drop of the packed bed versus particle Reynolds number. Measured values are plotted with Ergun's original equation, Eqn. (1), and the equation presented in Einfeld and Schnitzlein, Eqn. (5), which accounts for the wall-effect. The Sauter diameter was used in both equations. In calculating Eqn. (5), the coefficients for "all shapes" were used, as defined in Table 1. Both correlations are plotted with Singh et al.'s equation in Fig. 3, which uses the equivalent particle diameter by volume. Singh's equation shows a relatively good fit in the Reynolds numbers

$$H = 18 + 8 \times 10^{-3} Q \left(\frac{m^3}{h} \right) - 1,5 \times 10^{-2} Q^2 \left(\frac{m^3}{h} \right)$$

$$H_1 = H_2 \left(\frac{N_1}{N_2} \right)^2 \quad Q_1 = Q_2 \frac{N_1}{N_2}$$

$$H_2 \left(\frac{N_1}{N_2} \right)^2 = 18 + 8 \times 10^{-3} Q_2 \frac{N_1}{N_2} - 1,5 \times 10^{-2} Q_2^2 \left(\frac{N_1}{N_2} \right)^2$$

$$H_2 = 18 \cdot \left(\frac{N_2}{N_1} \right)^2 + 8 \times 10^{-3} \frac{N_2}{N_1} Q_2 - 1,5 \times 10^{-2} Q_2^2$$

$$N_1 = 2900 \quad N_2 = N_{min}$$

$$H_2 = 2,14 \times 10^{-6} N_2^2 + 2,76 \times 10^{-6} Q_2 N_2 - 1,5 \times 10^{-2} Q_2^2$$

$$H_2 = 10,76 \text{ m} \quad Q_2 = 4,38 \text{ m}^3/h$$

$$0 = 2,14 \times 10^{-6} N_2^2 + 1,208 \times 10^{-5} N_2 - 11,05$$

$$\Rightarrow \boxed{N = 2270 \text{ rpm}}$$

$$c) \quad P_E = \frac{P_M}{\eta_H} \quad P_{M2} = P_{M1} \left(\frac{2270}{2900} \right)^3$$

Determino homólogo por conocer P_{M1}

$$Q_1 = 4,38 \cdot \frac{2900}{2270} \Rightarrow Q_1 = 5,60 \text{ m}^3/h \Rightarrow P_{M1} = 0,59 \text{ kW}$$

$$\Rightarrow P_M = 0,28 \text{ kW} \Rightarrow P_E = 0,33 \text{ kW}$$

$$EE = P_E \cdot \Delta t (h) \Rightarrow \boxed{EE = 8,0 \text{ kW.h}}$$



difference between the tank diameter and the bed diameter is due to a 0.152 m thick flexible ceramic wool insulation that lines the tank and which will be used for future heat transfer experiments. To estimate the diameter of the bed, an average value was taken by measuring the diameter at various positions. The uncertainty of this measurement was included in the analysis of the air mass flux. A 9.53 mm thick perforated carbon steel plate with 12.7 mm holes and 48% open area is used to support the pellets. The upper and lower elbows are 0.161 m in diameter. A diffuser plate was installed at the exit of the lower elbow to reduce the effects of flow separation in the elbow, which creates non-uniform flow conditions. Room temperature (23°C) air entered the bottom of the system through one of 3 blowers that were used to achieve the desired flow rates. Velocity was measured with a hot wire anemometer at a position that had at least 10 pipe diameters upstream and 4 pipe diameters downstream from the point of measurement. Pressure was measured with a digital manometer. Specifications and accuracy values for the instruments are provided in Table 2.

Table 2. System components.

Instrument	Description	Accuracy
Hot wire anemometer	0.15 – 30 m/s range	± 3% reading
Digital manometer	0 – 1 in. w.c.(0 – 249.1 Pa) range	± 0.5% F.S./ ± 0.1% F.S. hysteresis
Digital manometer	0 – 20 in. w.c.(0 – 4,982 Pa) range	± 0.5% F.S./ ± 0.1% F.S. hysteresis
Dayton Blower	¼ hp	-
Dayton Blower	1/30 hp	-
Heavy Duty Blower	7.5 hp / 5.5 kW, 3 phase induction motor	-

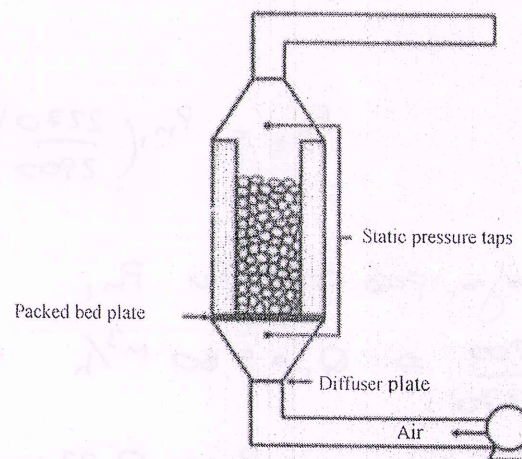


Fig. 1. System schematic.

To calculate bed porosity, the mass of the pellets and bed volume were used to estimate the bulk density which was then used in the following equation to calculate porosity:

$$\varepsilon = 1 - \frac{\text{Bulk density}}{\text{True density}} \quad (13)$$

Porosity was also measured in a separate container with a diameter that was similar to the final bed diameter to verify the voidage.

The equivalent diameter of the pellets was obtained by measuring the mass of 35 random samples. The volume of each of the samples was then calculated using the true density, and the average volume of the 35 samples was used to obtain the equivalent particle diameter by volume as defined in Eqn. (10) with a standard deviation of 0.006426 m. The bed parameters are listed in Table 2.

1) d)



(A, J) Velocidad nominal

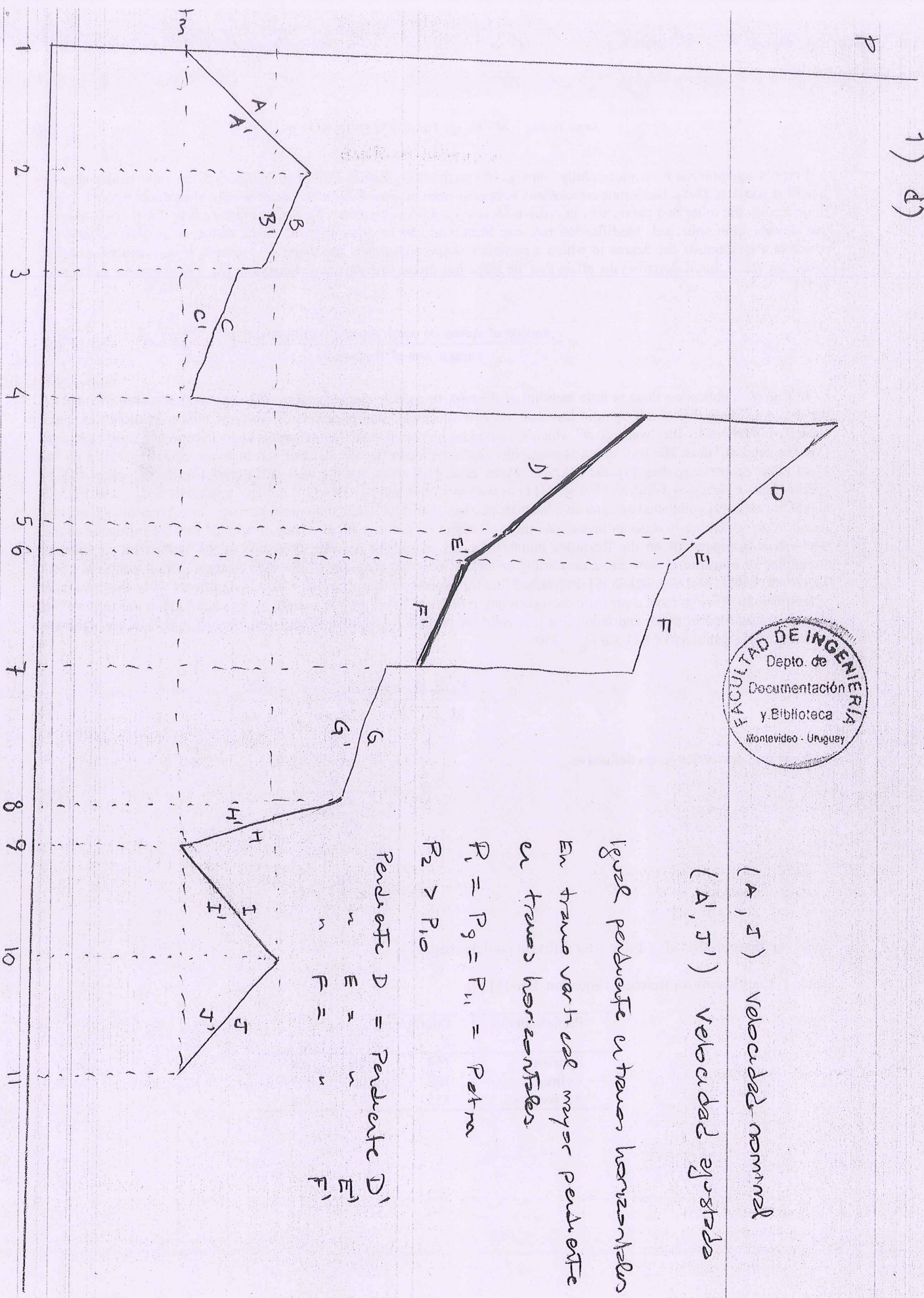
(A', J') Velocidad ajustada

Igual pendiente a tramos horizontales
En tramo variable mayor pendiente que
a tramos horizontales

$$P_1 = P_g = P_{11} = P_{atm}$$

$$P_2 > P_{10}$$

Pendiente D = Pendiente D'
" E = "
" F = "
" E' = "
" F' = "



Ergun's equation has been successfully employed to predict the pressure drop of packed beds filled with regular-shaped spherical particles [3-5]. Numerous correlations have also been proposed, some of which simply alter the constants A and B , or modify the other bed parameters in order to develop a more accurate prediction of pressure drop for particle shapes that deviate from spherical. Modification has also been made by incorporating the shape factor, or particle sphericity, which is a measure of the degree to which a particle's shape approaches the shape of a sphere. It does not necessarily define the shape, but it describes the effect that the shape has on the hydrodynamic behaviour [6]. The sphericity is defined as [7]

$$\psi = \frac{\text{surface of sphere of equal volume to the particle}}{\text{surface area of the particle}} = \frac{\pi^{1/3} (6V_p)^{2/3}}{A_p} \quad (4)$$

In Ergun's publication there is little mention of the tank-to-particle diameter ratio, d/D_p , except that scatter in some of the data was likely due to a d/D_p ratio less than 10. It is speculated that packed beds with low tank-to-particle ratios are strongly influenced by the "wall-effect" which is caused by increased void fraction and viscous friction at a rigid tank wall [8]. The ratio at which the wall effect is negligible has been inconsistently defined and there are opposing views on its exact effect on pressure drop [9]. For example, Meier et al. [10] claim that the wall-effect can be neglected when d/D_p is greater than 40, whereas Torab and Beasley [11] as well as Cohen and Metzner [12] indicate a value of $d/D_p > 30$. Eissfeld and Schnitzlein [9] conducted an extensive analysis of more than 2300 data points from pressure drop experiments, mostly found in the literature, in order to investigate the wall-effect on pressure losses. They concluded that the influence of the wall-effect is dependent on the Reynolds number, i.e. an increasing pressure drop due to the wall-effect appears in streamline or transitional flow and a decreasing or lack of pressure drop due to the wall emerges during turbulent flow. Moreover, Eissfeld and Schnitzlein [9] determined that for streamline flow, the wall-effect is significant when the d/D_p ratio is less than 10. They present a pressure drop correlation that was developed in a study by Reichelt [13] in the form of the dimensionless friction factor and found that it is valid for particle Reynolds numbers of $0.01 \leq Re_p \leq 17635$, and tank-to-particle diameter ratios of $1.624 \leq d/D_p \leq 250$:

$$f = \frac{K_1 A_w^2 (1-\varepsilon)^2}{Re_p \varepsilon^3} + \frac{A_w (1-\varepsilon)}{B_w \varepsilon^3} = \frac{\Delta P D_p}{\rho_f L U^2} \quad (5)$$

with the wall correction terms defined as

$$A_w = 1 + \frac{2}{3 \left(\frac{d}{D_p} \right) (1-\varepsilon)} \quad (6)$$

$$B_w = \left[k_1 \left(\frac{D_p}{d} \right)^2 + k_2 \right]^2 \quad (7)$$

and k coefficients defined in Table 1 for different particle shapes.

Table 1. Coefficients for Reichelt's equation, Eqn (5) [9].

Particle shape	Coefficients		
	K_1	k_1	k_2
Spheres	154	1.15	0.87
Cylinders	190	2.00	0.77
All particles	155	1.42	0.83

2) (b) Modelo adimensional (flujo sónico) $\left\{ \begin{array}{l} \rightarrow \text{Estimo y Laplace} \\ \text{Flujo desde reservorio} \end{array} \right.$
 Assum RCT# $\Rightarrow f_s = 0,029 \Rightarrow N = 17,5$

con $N = 17,5 \Rightarrow P_3/P_0 \approx 0,16 \Rightarrow \boxed{P_{\text{máx. en la garganta}} = 0,16 \text{ bar}}$
 # (Prácticamente no cambia si uso $N = 19,5$)
 por el caso flujo sónico

(c) Simplificaciones bajo condiciones sónicas.

$$\left(\frac{\gamma+1}{2\gamma} \right) x = z^2$$

$$U = \frac{2\gamma r^2}{(\gamma+1)x}$$

$$(2AS)_2 \quad U = \frac{8\gamma f L_c' + 1 + h U}{(\gamma+1) D} \Rightarrow U = 24,6$$

$$(1'') \quad r = \left(1 - \frac{(\gamma-1)}{(\gamma+1)U} \right)^{\frac{1}{(\gamma-1)}}$$

$$x = \gamma z F \Rightarrow x = 1,4 \cdot z \cdot 0,16$$

$$x = 4,58 \times 10^{-2}$$

$$\Rightarrow z = 0,204 \Rightarrow v_2 = 4,08 \frac{\text{m}^3}{\text{kg}}$$

$$\omega = \frac{u \cdot A}{V} \Rightarrow u = \frac{\omega}{A} \cdot V \Rightarrow u_2 = 302 \text{ m/s}$$

$$\# Re = \frac{\omega}{A} \frac{D}{\mu} = 4,2 \times 10^4 \left\{ \begin{array}{l} \Rightarrow f_s = 0,032 \Rightarrow 4f \frac{L}{D} = 18,5 \\ e/D = 4,4 \times 10^{-3} \end{array} \right.$$

$$\Rightarrow \text{De } (2AS)_2 : U = 25,8$$

$$\Rightarrow \text{De } (1'') : r = 0,9839 \quad \left| \begin{array}{l} \Rightarrow \\ \text{Def. U} \end{array} \right.$$

$$x = 4,37 \times 10^{-2} \Rightarrow \frac{\omega}{A} = 72,4 \frac{\text{kg}}{\text{m}^2}$$

$$x = \gamma z F \Rightarrow x = 1,4 z \cdot 0,16 \left\{ \begin{array}{l} \Rightarrow z = 0,195 \Rightarrow v_2 = 4,27 \frac{\text{m}^3}{\text{kg}} \\ x = 4,37 \times 10^{-2} \end{array} \right.$$

$$\omega = \frac{uA}{V} \Rightarrow u = \frac{\omega}{A} \cdot V \Rightarrow \boxed{u_2 = 309 \text{ m/s}}$$



Table 3. Percent average relative absolute error (ARAE) for the 3 pressure drop correlations in which sphericity was considered.

Correlation	% ARAE
Ergun	6.9%
Eisfeld and Schnitzlein	10.1%
Singh et al.	14.2%

An attempt was also made to calculate the sphericity using the equivalent particle diameter defined by Li and Ma [4], however the sphericity that was found with this method did not produce reasonable results in Singh et al.'s equation. Figure 5 shows the results of the predictions of the three correlations if the shape factor is determined by the use of Li and Ma's equivalent particle diameter, D_{eq} . This results in a shape factor value of 0.70. As can be seen in the figure, Singh et al.'s equation severely underpredicts the pressure drop when the higher shape factor is introduced. The percent average relative absolute error for Ergun's equation in this instance is 7.6% and for Eisfeld and Schnitzlein is 7.5%. For Singh et al.'s correlation it is 43.9%.

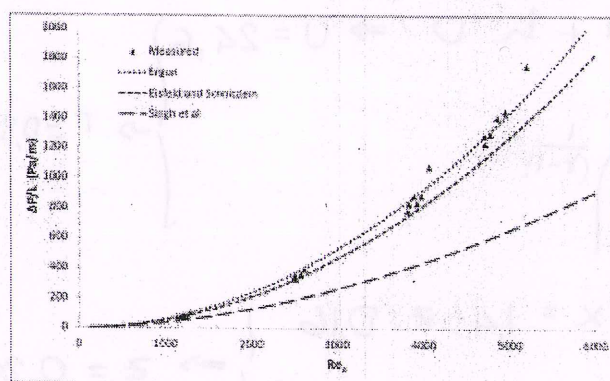


Fig. 5. Pressure gradient as a function of particle Reynolds number. Shape factor of 0.70 is used in the pressure drop correlations.

4. Conclusions

The purpose of this study was to find a pressure drop correlation that can reasonably predict pressure losses in a bed of highly irregular shaped solid pellets. Many correlations are available in the literature and there is little agreement on any universal method that can be used to accurately predict the pressure gradient for non-spherical shapes. There has even been some dispute in the accuracy of Ergun's correlation for the prediction of spherical particles [14]. With the many factors that can come into play, it is of no surprise that there is considerable discrepancy between correlations. Key parameters such as inlet and outlet conditions, tank-to-particle ratio, packing method, bed porosity, and particle shape can affect the studies. In [15] it was found that by simply washing aggregates of crushed stone and rock, pressure losses decreased by a factor of approximately 2. Correlations are also highly sensitive to slight changes in porosity and particle diameter, which can often be difficult to measure accurately.

The equivalent particle diameter by volume, D_p , is the simplest diameter to determine yet it does not necessarily provide a complete picture of the hydraulic behaviour of the bed. The Sauter diameter, which is the product of the sphericity and D_p , is known to be a more appropriate property, however the sphericity is very difficult to measure for